BLUE FAIRY DEVELOPMENT

# BlueFairy Brew Tracker

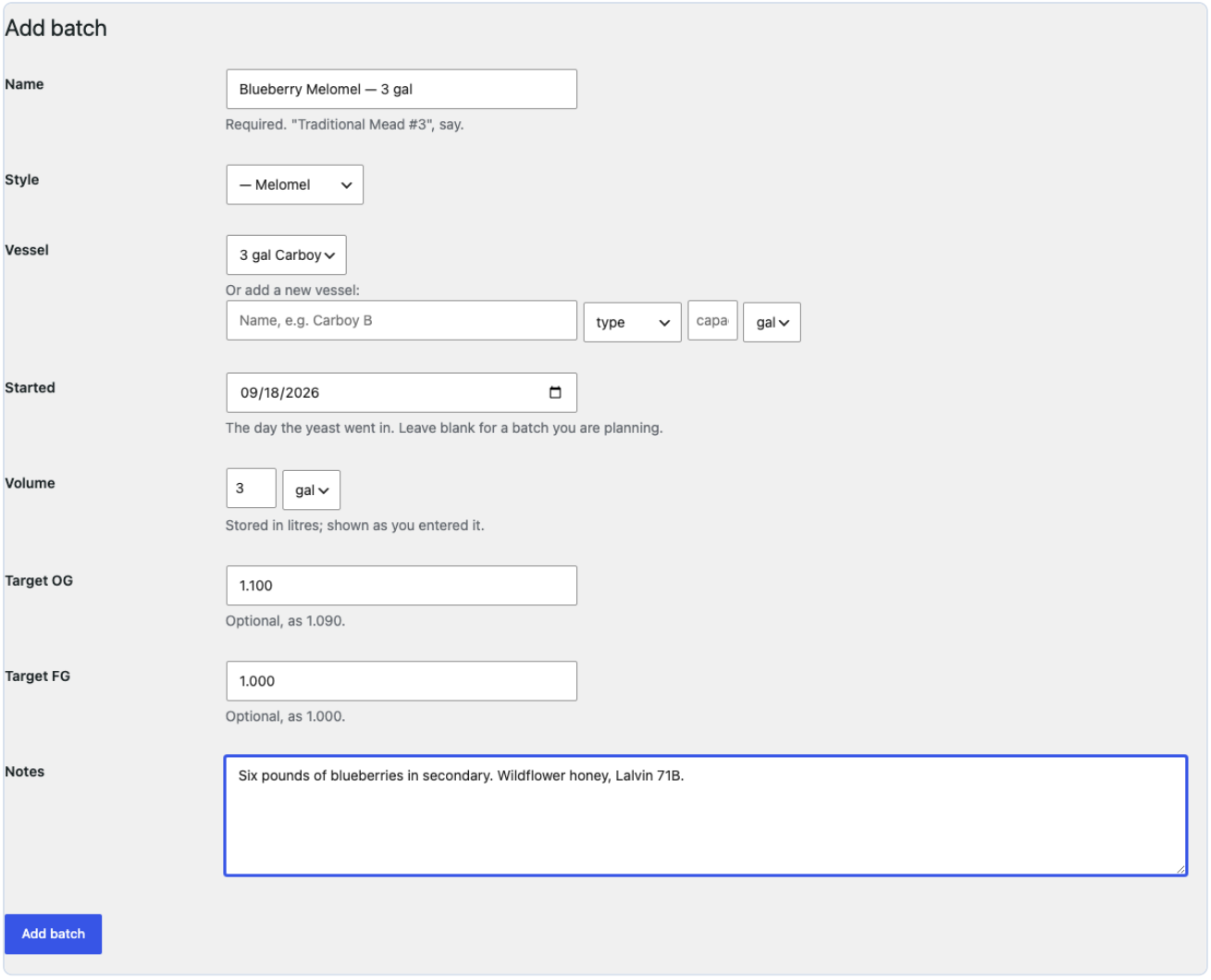
A batch log for mead, wine, cider and beer — gravity, ABV, timeline, tasks

Version 1.0.0 · GPL-2.0-or-later · WordPress 6.6+ · PHP 8.1+
bluefairydevelopment.com

# BlueFairy Brew Tracker — features

Every statement here is a claim about the shipped code. If one stops being true, the code is wrong or this file is.

## The batch record




Adding a batch: name, style, vessel, start date, volume, target OG and FG

One record, `bfbt_batch`, owned by this plugin. The Pro add-on extends the same record through `BFBT_Batch::register_field()` rather than keeping its own, which is why installing Pro shows you every batch you have already logged.

| Field | Where it lives |
|---|---|
| Name | the post title |
| Style | a term in the `bfbt_style` taxonomy |
| Vessel | a term in the `bfbt_vessel` taxonomy, with its type and capacity as term meta |
| Started | `_bfbt_started` — the day the yeast went in; blank for a batch you are planning |
| Volume | `_bfbt_volume_l` in litres, with `_bfbt_volume_entered` keeping "1 gal" as you typed it |
| Target OG, target FG | `_bfbt_target_og` , `_bfbt_target_fg` |
| Racked to secondary, packaged, completed | `_bfbt_secondary_date` , `_bfbt_packaged_date` , `_bfbt_completed_date` |
| Dumped | `_bfbt_dumped` — `1` , or absent |
| Original gravity | `_bfbt_og_log_id` — which reading counts as OG |
| Notes | the post content, through `wp_kses_post` |

The style tree is seeded on activation: Mead (Traditional, Melomel, Cyser, Metheglin, Braggot, Pyment), Wine (Red, White, Rosé, Fruit, Sparkling), Cider (Dry, Sweet, Perry), Beer (Ale, Lager, Stout, IPA, Wheat, Sour) — twenty-four terms. Seeding is by slug and never overwrites a term, so activating twice seeds once and a style you renamed keeps its name.

A vessel is a term with a type (carboy, bucket, demijohn, conical, keg, barrel, other) and a capacity. You pick an existing vessel from a list or type a new one on the batch form, so "Carboy A" stays one vessel however many batches go through it.

**Check your batches from the command line:**

```
 wp post list --post_type=bfbt_batch --fields=ID,post_title,post_status
wp post list --post_type=bfbt_log --post_parent=<batch id> --fields=ID,post_title,post_date
wp post list --post_type=bfbt_task --post_parent=<batch id> --fields=ID,post_title
wp term list bfbt_style --fields=term_id,name,slug,parent
wp term list bfbt_vessel --fields=term_id,name
```

## Validation refuses, never guesses

Every value is checked before anything is written and refused with the value quoted back. A refused save is never reported as a success.

- **Dates** are calendar-checked in year-month-day. `2026-02-30` has the right shape and is not a day, so it is refused. `03/14/2026` is refused rather than read as the 3rd of February.
- **Gravity** must be typed as `1.090` , in `0.980` to `1.200` . `1090` , `90` and a value that looks like Brix are refused with the hint "enter as 1.090" rather than reinterpreted into a reading you did not take.

- **Quantities** are non-negative decimals with a dot. **Units** come from a fixed list: g, kg, oz, lb, ml, l, gal, tsp, tbsp, pkt, tablet, each.
- **Temperature** is °C in –10 to 60; **pH** is 0 to 14.
- **Volume** is a number and a volume unit (ml, l, gal) and must be more than zero.

## Units

Volumes are stored in litres, mass in grams, temperature in °C. One setting, **Display units** (US or metric), decides how a *derived* volume is shown — the batch size, say, appears as "1 gal (3.79 l)" in metric and "1 gal (1.00 gal)" in US. A quantity you logged is never converted: a 1.2 g Fermaid-O dose is shown as 1.2 g whatever the setting says.

## The timeline

Every entry is a `bfbt_log` post with the batch as its parent, dated when it happened rather than when it was typed (the "When" field, `Y-m-d H:i`, calendar-checked). Seven entry types: addition, gravity, temperature, pH, note, racking, stage.

| Type | What it records | Generated title |
|---|---|---|
| Addition | ingredient as typed, quantity, unit | `Fermaid-O 1.2 g` |
| Gravity | SG, and a temperature if you took one | `SG 1.042 @ 20 °C` |
| Temperature | °C | `20 °C` |
| pH | pH | `pH 3.40` |
| Racking | optionally the volume that came across | `Racked — 3.5 l across` |
| Note | free text | the first sixty characters |
| Stage | a label | the label |

**Entries are immutable.** There is no edit form and the handler refuses any posted entry id. A typo is trashed and re-entered, so a number on the batch screen can always be traced to the entry that produced it.

The timeline shows fifty entries a page, newest first. Trash is offered only to users who may delete others' posts.

An entry dated ahead of the site clock stays published. WordPress would otherwise schedule it as a future post and drop it from every query.

## Original gravity, ABV and attenuation

# Traditional Mead #3 Secondary Edit All batches

Traditional · Carboy A · 1 gal (1.00 gal) · started 2026-08-04 · 20 days in Secondary

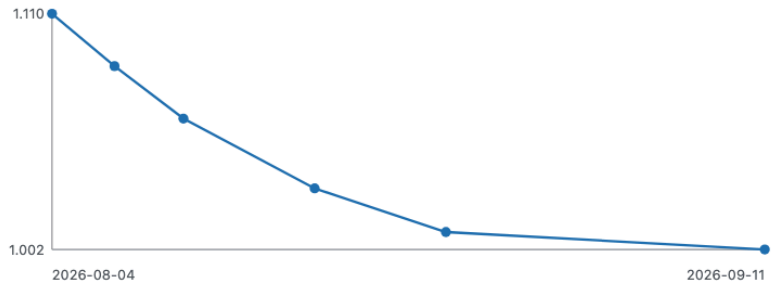
**ABV 15.6 %** · apparent attenuation 98 %

OG 1.110, reading of 2026-08-04 · FG 1.002, reading of 2026-09-11

76.08 × (OG − FG) / (1.775 − OG) × (FG / 0.794), the high-gravity formula for OG above 1.070

Orange blossom honey, 3 lb to the gallon. Staggered Fermaid-O, degassed twice daily for the first week.

## Gravity




## Due

| | | |
|---|---|---|
| 2026-09-16 | Gravity check — confirm stable | Done |
| 2026-09-19 | Potassium metabisulphite 1 tablet | Done |

### Done in the last 7 days

| | | |
|---|---|---|
| 2026-09-18 | ~~Potassium sorbate 0.5 tsp~~ | Undo |

## Add entry

Type: Addition · When: 2026-09-18 16:57

Ingredient: Fermaid-O

Quantity: g

Note

Add entry

## Add task

Type: Addition

Ingredient: Fermaid-O

Quantity: g

Leave the quantity blank and put the instruction in the title — "water to 1 gal".

Title (optional)

Due: mm/dd/yyyy — Blank for a task with no date.

Add task

## Timeline

| When | Type | Entry | Note | |
|---|---|---|---|---|
| 2026-09-18 16:56 | Addition | Potassium sorbate 0.5 tsp | | Trash |
| 2026-09-11 10:00 | Gravity | SG 1.002 @ 19 °C | | Use as OG Trash |
| 2026-08-29 11:00 | Racking | Racked — 3.5 l across | Racked off the lees into a clean carboy. | Trash |
| 2026-08-25 10:00 | Gravity | SG 1.010 | | Use as OG Trash |
| 2026-08-18 10:05 | pH | pH 3.40 | | Trash |
| 2026-08-18 10:00 | Gravity | SG 1.030 @ 20 °C | | Use as OG Trash |
| 2026-08-14 08:00 | Temperature | 20 °C | | Trash |
| 2026-08-11 10:10 | Gravity | SG 1.062 | | Use as OG Trash |
| 2026-08-11 10:00 | Addition | Fermaid-O 1.2 g | | Trash |
| 2026-08-07 18:00 | Gravity | SG 1.086 @ 22 °C | | Use as OG Trash |
| 2026-08-06 10:00 | Addition | Fermaid-O 1.2 g | | Trash |
| 2026-08-05 10:00 | Addition | Fermaid-O 1.2 g | | Trash |
| 2026-08-04 10:05 | Addition | Fermaid-O 1.2 g | | Trash |
| 2026-08-04 10:00 | Gravity | SG 1.110 @ 21 °C | | OG Trash |
| 2026-08-04 09:45 | Addition | Lalvin 71B 1 pkt | | Trash |
| 2026-08-04 09:30 | Addition | Orange blossom honey 3 lb | | Trash |

## Milestones

Racked to secondary: 08/29/2026

Packaged: mm/dd/yyyy

Completed: mm/dd/yyyy

☐ Dumped

Save milestones

The first gravity reading you log is pinned as the original gravity. Any other gravity reading on the timeline has a **Use as OG** button; trashing the pinned reading clears the pin.

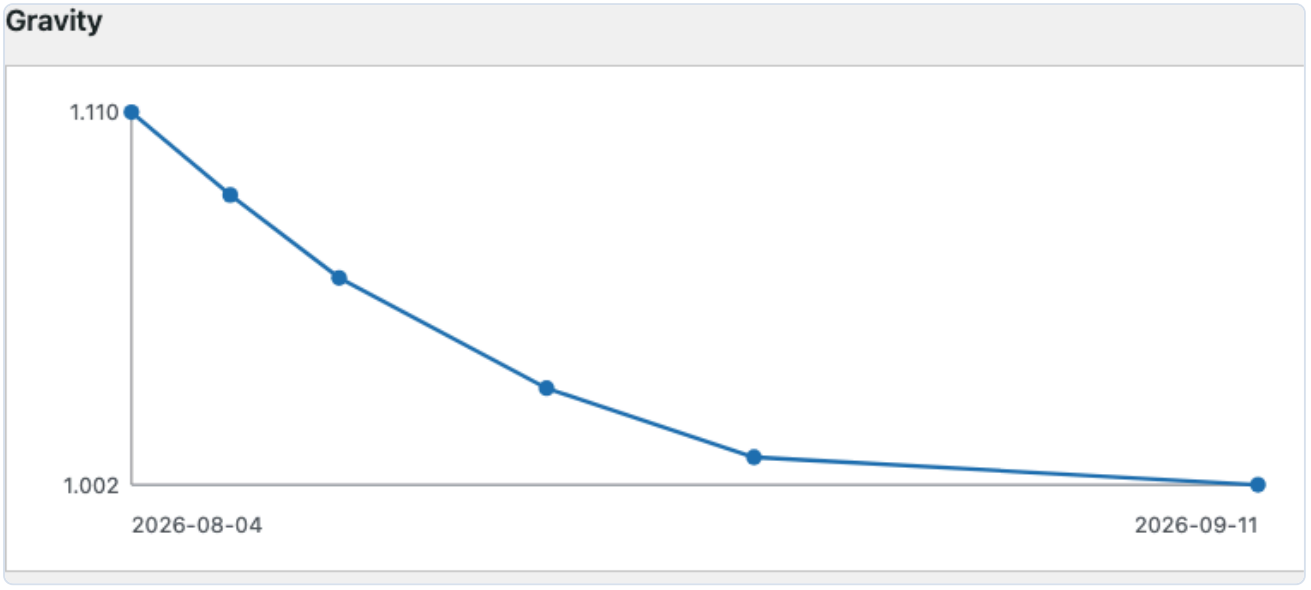
ABV and apparent attenuation are worked out at read time from the pinned OG and the latest gravity reading. Neither number is stored, and both are shown with their provenance — "OG 1.110, reading of 2026-03-01 · FG 1.000, reading of 2026-03-21" — so trashing or re-dating an entry changes a visible number, never a hidden one.

Two formulas, and the screen names the one it used:

| OG | Formula |
|---|---|
| up to 1.070 | `(OG − FG) × 131.25` |
| above 1.070 | `76.08 × (OG − FG) / (1.775 − OG) × (FG / 0.794)` |

The standard formula under-reports a mead by around 1.4 % ABV, which is most of the difference between a session mead and a sack mead.

## The gravity chart




Every gravity reading over time, drawn by the plugin as inline SVG

An inline SVG of every gravity reading over time, up to five hundred points, drawn by the plugin with no JavaScript library and no external asset. Hovering a point shows the reading and its date. Every number is cast to a float and every label is escaped before it reaches the markup. A batch with no readings says "No readings yet."

## Stage

Derived from the milestone dates, never stored, so a batch cannot disagree with itself.

| Stage | When |
|---|---|
| Dumped | the batch is marked dumped |
| Complete | a completed date is recorded |
| Packaged | a packaged date is recorded |
| Secondary | a racked-to-secondary date is recorded |
| Primary | a start date is recorded |
| Planned | nothing yet |

Most final wins: a batch that was completed and then poured away is dumped, and the completed date left behind does not hide that. The batch screen says how many days it has been in its stage.

Pro replaces the answer through the `bfbt_batch_stage` filter and inserts its own states into the order through `bfbt_stage_order`. It supplies only the states between the ends; planned, complete and dumped are decided here, once.

## Tasks and what is due

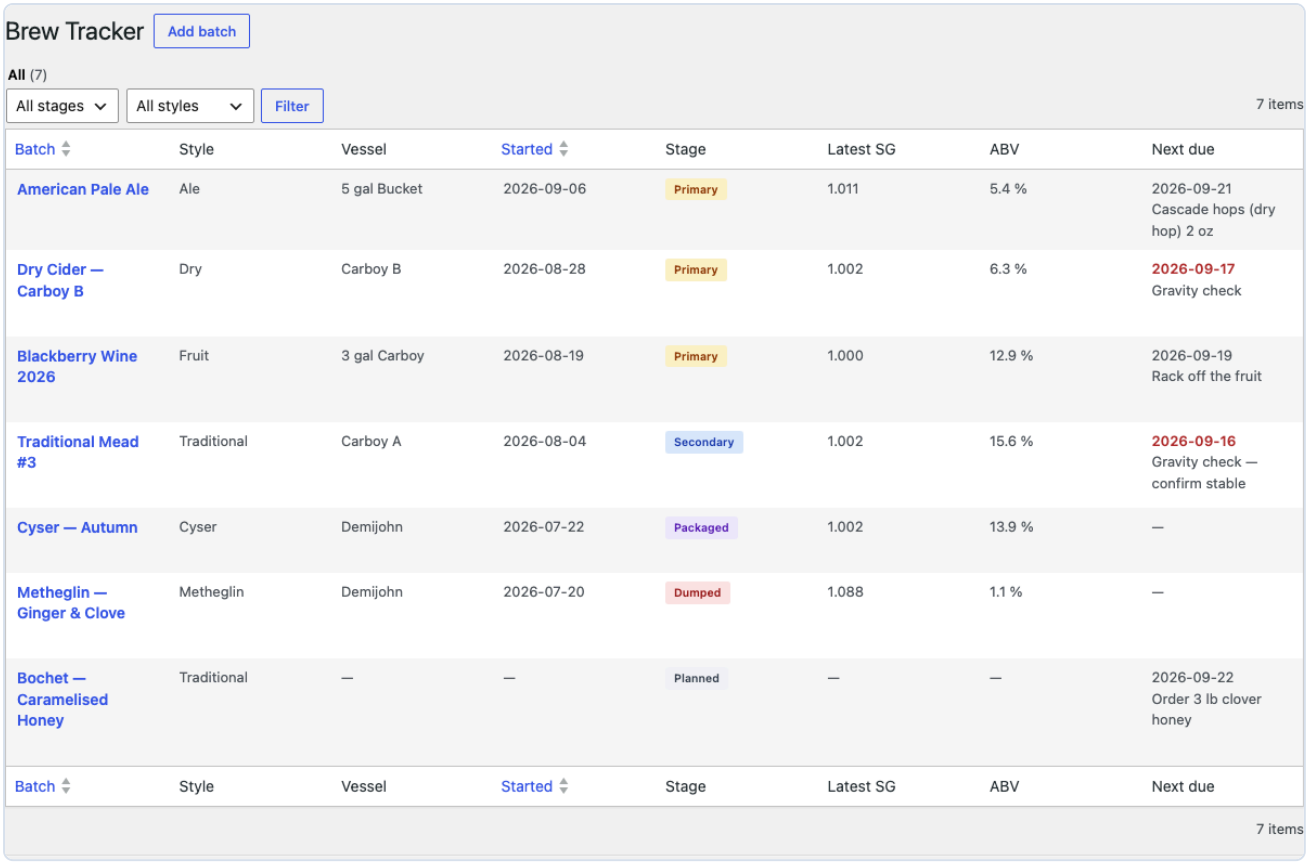
A task is a `bfbt_task` post: what to do, when, and when it was done. Five types: addition, gravity check, rack, package, other. An addition task carries an ingredient, quantity and unit, so "Fermaid-O 1.2 g" is one line and one button; leave the quantity blank and the instruction lives in the title ("water to 1 gal").

The **Due** panel on the batch screen lists open tasks soonest first — overdue in red, today in amber, undated in grey. **Done** logs the addition (or a note "Done: rack") to the timeline and stamps the task; **Undo** reopens it and leaves the entry in place. Completing is idempotent: a double-click or replayed form cannot log the same task twice.

The **Next due** column on the batch list shows each batch's earliest open task, red when overdue, worked out in one grouped query for the whole page. A task with no date never becomes "next due".

Tasks are typed by hand. Nothing here generates a schedule; that is the first thing Pro adds.

## The list

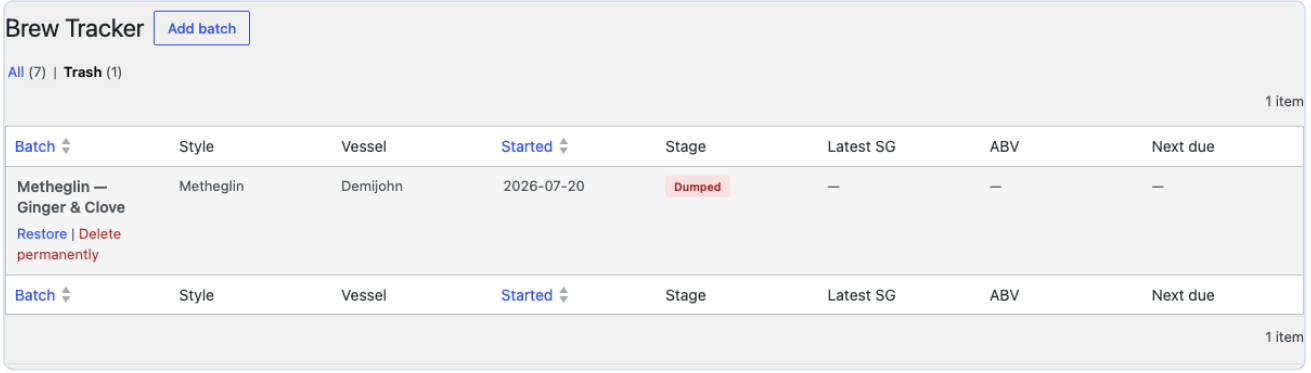



Every batch with its stage, latest gravity, ABV and what is due next

Twenty-five batches to a page, sorted and paged **in the database**. Columns: Batch, Style, Vessel, Started, Stage, Latest SG, ABV, Next due. Sort by name or by start date in either direction — a planned batch with no start date stays on the list. Filter by stage and by style.

**All** | **Trash** views. Trashing a batch takes its entries and tasks with it; restoring brings back exactly the ones that went with it, not an entry you had trashed on purpose beforehand; deleting permanently from the Trash view removes the batch and its children for good.

## Who may do what

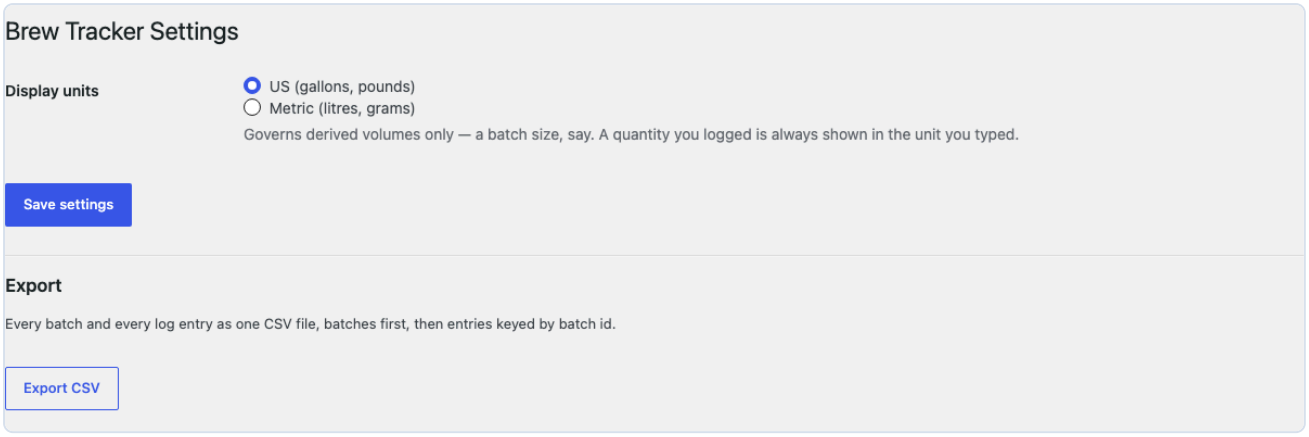



The Trash view: Restore brings back the batch with its entries and tasks; Delete permanently removes them for good

| Action | Capability |
| --- | --- |
| See the screen, add and edit batches, log entries, add and complete tasks, pin OG | `edit_posts` |
| Trash, restore and delete a batch; trash an entry | `delete_others_posts` |
| Settings and CSV export | `manage_options` |

A Contributor can log; only someone who may delete others' posts can destroy. Every write is a nonced form post to an `admin_post_*` handler that checks its capability first and confirms the post type of every id it is handed.

## CSV export




Settings: display units, and the CSV export of every batch and entry

**Settings** → **Export CSV** streams every batch and then every entry as one file, five hundred rows at a time, never built in memory. Every field is quoted, and a value beginning with `=`, `+`, `-`, `@`, a tab or a carriage return is prefixed with a single quote so a batch named `=HYPERLINK(...)` opens in a spreadsheet as text, not as a formula.

## Uninstall

Uninstalling deletes `bfbt_features`, `bfbt_units`, `bfbt_seeded`, the plugin's one-shot notice transients, and `bfbt_style_children` — the cached parent/child map WordPress itself keeps for the style taxonomy — never a batch, an entry, a task, a style or a vessel. There is no `LIKE 'bfbt_%'` sweep: that pattern would also match the Pro add-on's `bfbtp_` settings.

## Extension points

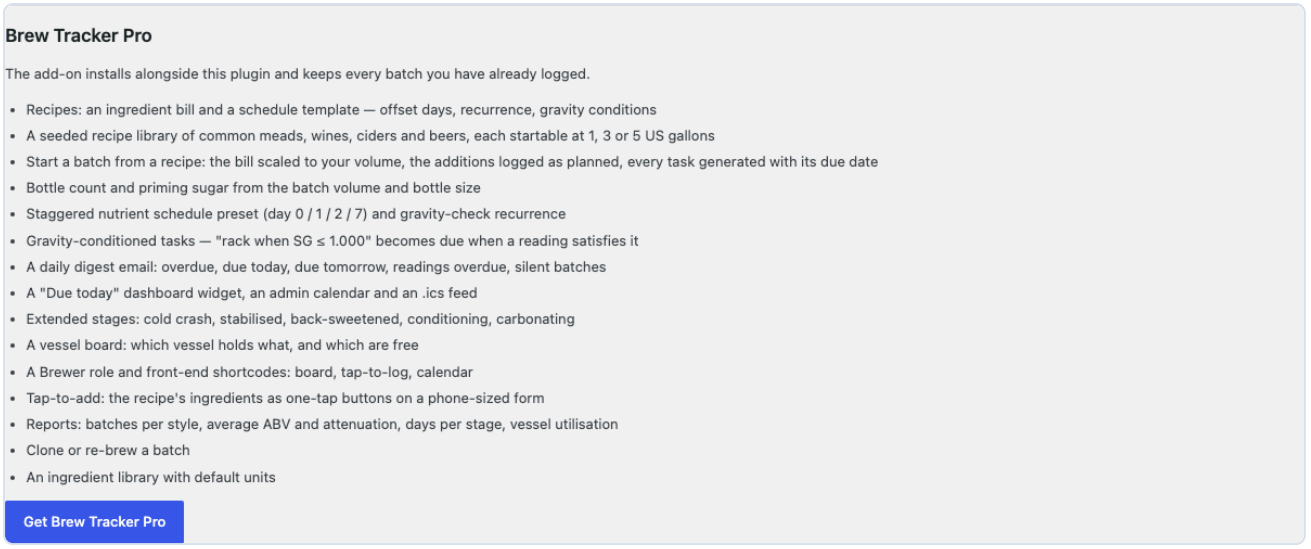
| Hook | Purpose |
|---|---|
| `bfbt_loaded` | fires before this plugin registers its own subsystems; where Pro attaches |
| `BFBT_Batch::register_field()` | add a field to the batch record |
| `bfbt_batch_stage`, `bfbt_stage_order`, `bfbt_terminal_stages` | replace the derived stage, insert states, name the stages after which nothing is scheduled |
| `bfbt_stage_labels`, `bfbt_stage_colours` | name and colour new states |
| `bfbt_log_types`, `bfbt_task_types` | add entry and task types |
| `bfbt_batch_created`, `bfbt_batch_updated`, `bfbt_log_created`, `bfbt_task_completed` | react to a change |
| `bfbt_batch_form_fields`, `bfbt_batch_form_submit` | add fields to the add/edit form and read them back |
| `bfbt_batch_actions` | add buttons to the batch screen |
| `bfbt_after_batch_list`, `bfbt_settings_after` | add content below the list and the settings |
| `bfbt_admin_menu` | add pages under the Brew Tracker menu |
| `bfbt_ingredient_label` | resolve an ingredient typed as text |
| `bfbt_units` | override the display-unit setting |
| `bfbt_feature_labels`, `bfbt_feature_descriptions` | declare feature switches |

Signatures are in `README.md`.

## What this plugin does not do

- No schedule. It works out ABV and attenuation from what you logged, and nothing else; every task is one you typed.
- No email, no cron, no scheduled work, no dashboard widget.
- No REST routes, no front end, no database tables of its own.
- Three options of its own, all removed on uninstall, along with the style hierarchy cache WordPress keeps for it.

## Free and Pro




What the paid add-on adds, shown beneath your batch list

This plugin is the base of a pair. Pro is an add-on: it extends this plugin rather than replacing it, so there is one batch record, one place your data lives, and nothing to migrate when you upgrade.

Pro column describes Brew Tracker Pro 1.0.

| | Pro — $79 | Free |
|---|---|---|
| Batch record: name, style, vessel, start date, volume, target OG/FG, notes | Yes | Yes |
| Style taxonomy seeded with Mead, Wine, Cider, Beer and their substyles | Yes | Yes |
| Vessel taxonomy with type and capacity | Yes | Yes |
| Timeline log per batch: additions, gravity, temperature, pH, notes, racking | Yes | Yes |
| ABV and attenuation from the pinned OG and the latest reading, both formulas stated | Yes | Yes |
| Gravity chart per batch, inline SVG | Yes | Yes |
| Six-stage cycle: planned, primary, secondary, packaged, complete, dumped | Yes, extended | Yes |
| Manual tasks with due dates, a Next due column and a Due panel | Yes | Yes |
| Paged, sortable batch list with a Trash view and Restore | Yes | Yes |
| CSV export of batches and entries | Yes | Yes |
| Batches kept on uninstall | Yes | Yes |
| **Recipes** — ingredient bill and schedule template | Yes | — |
| **Seeded recipe library** of common meads, wines, ciders and beers at 1, 3 or 5 US gallons | Yes | — |
| **Start a batch from a recipe** — bill scaled, additions planned, tasks generated | Yes | — |
| **Bottle count and priming sugar** from batch volume and bottle size | Yes | — |
| **Staggered nutrient preset** (day 0 / 1 / 2 / 7) and gravity-check recurrence | Yes | — |
| **Gravity-conditioned tasks** — "rack when SG ≤ 1.000" | Yes | — |
| **Daily digest email** — overdue, due today, due tomorrow, silent batches | Yes | — |
| **Dashboard widget, admin calendar and .ics feed** | Yes | — |
| **Extended stages** — cold crash, stabilised, back-sweetened, conditioning, carbonating | Yes | — |
| **Vessel board** — which vessel holds what, which are free | Yes | — |
| **Brewer role and front-end shortcodes** — board, tap-to-log, calendar | Yes | — |
| **Reports** — batches per style, average ABV, days per stage, vessel use | Yes | — |
| **Clone or re-brew a batch** | Yes | — |
| **Ingredient library** with default units | Yes | — |

## Why the missing rows cost you

**You are keeping the schedule in your head.** This plugin records the tasks you type. A one-gallon mead wants Fermaid-O on day 0, 1, 2 and 7, a gravity reading on day 3, 7 and 14, and a rack when the gravity settles — and you are the one counting the days for every carboy on the shelf. Pro keeps that as a recipe and "Start batch from recipe" creates every task with its date.

**Nothing tells you until you look.** The Due panel is on the batch screen, so you find out what is due when you open it. Pro sends a daily digest — overdue, due today, due tomorrow, batches that have gone quiet — and puts the same list on the dashboard, a calendar and a feed your phone subscribes to.

**A rack is a judgement, not a date.** "Rack to secondary when SG is at or below 1.000 and stable" cannot be typed as a task with a date. Pro's gravity-conditioned tasks become due when a reading satisfies them, and only when the last few readings agree.

**You are scaling the bill by hand.** Three pounds of honey for a gallon; how much for three? How many 16 oz bottles from five gallons after losses? Pro scales the recipe to the batch, keeps one yeast packet as one packet, rounds tablets up, and puts the bottle count on the packaging task.

**Nobody else can log a reading.** This plugin is a wp-admin screen. Pro adds a Brewer role and a phone-sized front end where the recipe's ingredients are one-tap buttons.

## Get Pro

**https://wordpress.bluefairydevelopment.com/product/bluefairy-brew-tracker-pro/**

One payment, no subscription, lifetime licence for one site. Pro requires this plugin to be installed and active — it is an add-on, not a replacement, and every batch you have already logged carries straight over.